Speaker Script — Cainiao is Fast, But Not Everywhere

Group 12 · Yingqi Tan · Hao Yang · Zhesheng Xu · BUAD 345 · Defense talk (~9–10 minutes).
One block per slide in `report/slides.pdf`. `[advance]` = move to the next slide. Numbers match `report/report.pdf` and the R output. The main visuals are built in **Tableau** (Cainiao = blue, non-Cainiao = orange); two composite **R dashboards** pull each half of the story onto a single screen.

Slide 1 — Title

Good [morning/afternoon], everyone. We're Group 12: Yingqi Tan, Hao Yang, and Zhesheng Xu. Our project asks a simple question with a not-so-simple answer: *Cainiao is fast — but is it fast everywhere?* We looked at 1.3 million Tmall orders from 2017 to find out. `[advance]`

Slide 2 — The question

Our audience is Cainiao itself, Alibaba's logistics platform. Why care about delivery speed? Because it pays. Fisher and colleagues found online sales rise about 1.45% for every business day you shave off delivery, and Rao and colleagues showed the flip side — miss a promised date and you lose loyal customers. Cainiao handles 30% of all the orders in our data, so if there's any pocket where its speed promise breaks, that matters a lot. So we asked: does Cainiao deliver faster — and does it do so everywhere? `[advance]`

Slide 3 — Method

Our outcome is total fulfillment time: when the customer signs for the package minus when they paid, in hours. The key to our approach is that we dig *hierarchically*, not in parallel. We start with Cainiao versus non-Cainiao, then add carrier size, then destination-city size, and finally we break the time into segments. Each layer is added only when the previous result makes us ask the next question. One focused story, not ten disconnected charts. `[advance]`

Slide 4 — Headline

Here's the headline, and it's good news for Cainiao. Cainiao orders take about 62 hours; non-Cainiao take 71 — Cainiao is roughly **nine hours faster**. And this isn't a trick of which orders Cainiao handles: if we keep only orders with no promised-speed window, the gap barely moves. So far, Cainiao's promise holds. `[advance]`

Slide 5 — Stable over time

And it holds *all the time*. This is daily average fulfillment time across the whole seven-month window. The blue line — Cainiao — sits below the orange line — non-Cainiao — every single month, including the busy spring. So the advantage is real and stable. That's what lets us pool the data and zoom in. `[advance]`

Slide 6 — The punchline

But here's where it gets interesting. When we split by both carrier size and city size, the picture cracks. In three of the four cells the blue Cainiao bar is shorter — Cainiao saves 8 to 10 hours. But look at the big-city, small-carrier cell — there the blue bar is *taller*: Cainiao is actually **3.3 hours slower** than non-Cainiao. The advantage doesn't just shrink; it reverses. `[advance]`

Slide 7 — The whole story on one screen

Before the numbers, let me show you the whole thing on one screen. Up top, the headline KPIs: 1.3 million orders, 30% on Cainiao, nine hours faster overall — but minus 3.3 in the problem cell. The 2×2 on the left and the hours-saved bars on the right both point to the same place: every cell is blue and positive *except* big-city, small-carrier, which dips into the red. If you take one picture away from this talk, take this one. `[advance]`

Slide 8 — The reversal cell in numbers

Now the numbers. Big city, large carrier: Cainiao saves nine and a half hours. Small city, either carrier: eight to ten and a half hours saved. But big city, small carrier: minus 3.3 hours. And this isn't a tiny corner of the data — that cell has almost 23,000 orders. It's a real, reliable reversal. `[advance]`

Slide 9 — Customers feel it

And customers notice. This is the logistics review score, and I've zoomed the axis in so you can see the gaps. That same big-city, small-carrier cell is Cainiao's *lowest* score anywhere — 4.76 — while the non-Cainiao option right next to it is the *highest* — 4.86. So this isn't just an operations curiosity; it's hurting the customer experience exactly where Cainiao underperforms. `[advance]`

Slide 10 — Mechanism

So *where* does the extra time come from? We split fulfillment into pickup, line-haul — that's transit between facilities — and the last mile. The last mile is small and basically the same everywhere, four to six hours. Cainiao's real edge is faster line-haul — for example, 38 hours versus 48 on a typical lane. But in the reversal cell that edge disappears: 27 hours versus 27. The lost advantage is a lost *line-haul* advantage — not the last mile. `[advance]`

Slide 11 — Mechanism, faceted

Here's that same mechanism broken out cell by cell. Notice the last-mile bars are short and nearly identical in every panel — that is *not* the problem. It's the line-haul bars that move, and they stay stubbornly high for the big-city, small-carrier Cainiao orders. That's the leg to fix. `[advance]`

Slide 12 — An honest non-result

Now, we want to be honest about one thing. The obvious guess is that small carriers make more stops. We checked — and it's *not* true. Small carriers actually pass through *fewer* cities than large ones. So the problem isn't the number of hops; it's the time spent on each leg. Reporting that keeps our explanation disciplined. `[advance]`

Slide 13 — Why

Why would that happen? Two reasons consistent with the evidence. First, big cities concentrate trunk-line volume — large carriers run dedicated, high-frequency line-haul into them, while small carriers wait to consolidate loads, which adds dwell time on exactly the lanes where customers expect speed. Second, small carriers likely can't execute Cainiao's one-size-fits-all smart routing — and the routing literature shows that ignoring carrier capability degrades performance. `[advance]`

Slide 14 — So what

So what should Cainiao do? Three moves. One: deepen ties with large carriers on big-city lanes, where speed actually shows up — while watching the concentration risk. Two: make smart routing carrier-capability-aware, so small carriers aren't handed routes they can't run. Three: pilot big-city consolidation hubs or parcel lockers for small-carrier shipments, to attack the line-haul penalty where it lives. And because the weakness is concentrated, the fix is cheap compared to a platform-wide change. `[advance]`

Slide 15 — Two-sentence summary

If you remember two sentences: First — Cainiao makes fulfillment about nine hours faster across the marketplace, and that's robust. Second — but that advantage disappears, and reverses, for big-city destinations on small carriers, with the bottleneck in line-haul, not the last mile. And a caveat: this is observational data, so we're showing strong association, not proven causation. `[advance]`

Slide 16 — Thank you

That's our story: Cainiao is fast — just not everywhere — and now we know exactly where to look. Thank you. We're happy to take questions.

Timing & delivery notes

- Target ~9–10 min; slides 6–11 (punchline → faceted mechanism) are the core — slow down there.
- Slides 7 and 11 are the two composite R dashboards — use them to *recap* a half of the story in one glance; don't read every bar, just point to the one cell that breaks.

- Hand-off cues: if presenting as a team, natural splits are 1–5 / 6–11 / 12–16.
- Likely Q&A: *Is it causal?* (no — selection into Cainiao, see slide 15); *Why two big cities only?* (top-2 signing cities by volume, a coarse proxy); *Do segments add up?* (directional only — missing milestones, different subsets); *Tableau or R?* (analysis + aggregation in R; the story visuals in Tableau, with two R composite dashboards as one-screen recaps).